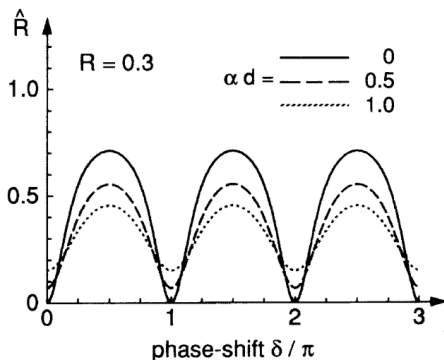


Homework for Chap 3

1. What are the definition and meaning of Einstein's coefficients?
2. Consider the following figure (Fig 3.9 (c) in the textbook), explain why the three curves intersect, i.e., why zero absorption gives both the highest maxima and the lowest minima in reflectance



1.

A_{21} : 自发辐射系数
 B_{21} : 受激辐射系数
 B_{12} : 吸收系数

$$\Rightarrow \frac{A_{21}}{B_{21}} = \frac{8\pi h \nu^3}{c^3}$$

$$B_{12} = B_{21}$$

2. $\alpha d = 0$ 对应零吸收

此时只存在干涉不存在吸收, 无能量损失, 故出现了最显著的干涉

在 $\delta = m\pi$ 时, 相长干涉, 反射率正到最低

$\delta = (m + \frac{1}{2})\pi$ 时, 相消, 达到最高

而 $\alpha d > 0$ 时, 存在能量被介质吸收, 能量在介质中消耗, 极大值更低, 极小值更大